MAYER-VIETORIS DISTINGUISHED TRIANGLE FOR DERIVED GLOBAL SECTIONS IN THE UNBOUNDED DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof defers exactness of the sequence of complexes to the proof of the Mayer-Vietoris lemma; the author's sentence is reproduced verbatim and that reference is carried as a navigation stub.

Editorial note. The counted spans carry 6 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-unbounded-mayer-vietoris is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- K-injective resolutions with injective terms, functorially. Injectives, Theorem injectives-theorem-K-injective-embedding-grothendieck, invoked at proof lines 8223-8225 and 8244-8246. Supplies the representative and the functoriality statement.
- Restriction to an open subspace preserves K-injectivity. Same chapter, Lemma lemma-restrict-K-injective-to-open, invoked at proof lines 8227-8228. Supplies that the restricted complexes compute the derived global sections over U, V and the intersection.
- Distinguished triangle attached to a short exact sequence of complexes. Derived Categories, Section derived-section-canonical-delta-functor and Lemma derived-lemma-derived-canonical-delta-functor, invoked at proof lines 8229-8234. Supplies the triangle once exactness of the displayed sequence is known.
- The termwise Mayer-Vietoris exact sequence of sections. Same chapter, Lemma lemma-mayer-vietoris, whose proof is cited for the exactness at proof lines 8242-8243. Supplies exactness of the sequence of complexes.

Source reference. Original lemma 0152: sources/stacks-chow/tags.txt, source lines 1493–1493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 014Z: sources/stacks-chow/tags.txt, source lines 1490–1490. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original theorem 079P: sources/stacks-chow/tags.txt, source lines 9279–9279. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EB: sources/stacks-cohomology/pinned-original-source.tex, source lines 767–767. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 08BS: sources/stacks-cohomology/pinned-original-source.tex, source lines 7840–7840. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Suppose that $X = U \cup V$ is a union of two open subsets. For an object E of $D(\mathcal{O}_X)$ we have a distinguished triangle*

$$R\Gamma(X, E) \to R\Gamma(U, E) \oplus R\Gamma(V, E) \to R\Gamma(U \cap V, E) \to R\Gamma(X, E)[1]$$

and in particular a long exact cohomology sequence

$$\ldots \to H^n(X, E) \to H^n(U, E) \oplus H^0(V, E) \to H^n(U \cap V, E) \to H^{n+1}(X, E) \to \ldots$$

The construction of the distinguished triangle and the long exact sequence is functorial in E.

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ representing E. We may assume $\mathcal{I}^n$ is an injective object of $Mod(\mathcal{O}_X)$ for all n, see Injectives, Theorem 079P. Then $R\Gamma(X, E)$ is computed by $\Gamma(X, \mathcal{I}^\bullet)$. Similarly for U, V, and $U \cap V$ by Lemma 08BS. Hence the distinguished triangle of the lemma is the distinguished triangle associated (by Derived Categories, Section 014Z and especially Lemma 0152) to the short exact sequence of complexes

$$0 \to \mathcal{I}^\bullet(X) \to \mathcal{I}^\bullet(U) \oplus \mathcal{I}^\bullet(V) \to \mathcal{I}^\bullet(U \cap V) \to 0.$$

We have seen this is a short exact sequence in the proof of Lemma 01EB. The final statement follows from the functoriality of the construction in Injectives, Theorem 079P. □